

Project Bank

WEB DEVELOPMENT PROJECT BANK

[HTML & CSS](#) • [JavaScript](#) • [HTML5 Application Development](#)

Starter Skills → Power-Up Skills → Legendary Skills

Each project includes a prompt, skills covered, and optional creative and challenge extensions. Feel free to put your own twist on any of these projects (or come up with your own project, using these for inspiration).

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HTML + CSS

HTML & CSS PROJECTS

Starter ★ (10 pts)

1. Personal Profile Page

Project Prompt

Create a one-page website that introduces *you* to the world. Think of it like your character sheet in a game — your story, your strengths, your interests, and what makes you unique.

What You'll Make

A clean, well-structured webpage with your name, photo, bio, and links.

Core Skills You'll Practice

- Structuring content with HTML
- Using headings, paragraphs, images, and links
- Styling with CSS (colors, fonts, spacing)

Creative Twist

Turn yourself into a game character with stats like Creativity, Coding Skill, Strength, or Luck.

Challenge Mode

Make the layout responsive using Flexbox so it looks great on phones.

2. Favorite Things Gallery

Project Prompt

Build a visual gallery that shows off your top interests — games, foods, hobbies, characters, anything you love.

What You'll Make

A grid or list of images with captions and descriptions.

Core Skills You'll Practice

- Displaying images
- Creating lists and sections
- Styling with color palettes and spacing

Creative Twist

Make each item a “collectible card” with rarity colors (Common, Rare, Epic).

Challenge Mode

Add hover animations or transitions to make the cards feel interactive.

3. Simple Portfolio

Project Prompt

Create a simple portfolio that showcases 3–4 projects, hobbies, or creations you’re proud of.

What You’ll Make

A multi-section webpage with project descriptions and links.

Core Skills You’ll Practice

- Multi-section layout
- Navigation links
- Consistent styling across a page

Creative Twist

Present each project as a “quest” with objectives and rewards.

Challenge Mode

Add a navigation bar that stays consistent across multiple pages.

Power-Up ★★ (15 pts)

4. Restaurant Menu Page

Project Prompt

Design a menu for a restaurant — real or fictional — with categories, prices, and images.

What You’ll Make

A visually organized menu with sections like Appetizers, Entrees, Desserts, etc.

Core Skills You’ll Practice

- Organizing content into categories
- Using CSS Grid for layout
- Typography and spacing

Creative Twist

Make it a fantasy tavern, sci-fi café, or themed restaurant.

Challenge Mode

Add responsive breakpoints so the menu rearranges on smaller screens.

5. Event Flyer Webpage

Project Prompt

Create a digital flyer that advertises an upcoming event. Think of it like a poster you'd see in a game hub — bold visuals, clear details, and a strong theme.

What You'll Make

A single-page promotional flyer with a title, date/time, description, and visuals that match the event's theme.

Core Skills You'll Practice

- Typography hierarchy
- Layout and spacing
- Using images and decorative elements
- Color theory for attention-grabbing design

Creative Twist

Make the flyer for an in-game event — a boss raid, tournament, festival, or faction meetup.

Challenge Mode

Add a styled RSVP form with input fields and a submit button. For an *added challenge*, try to add JavaScript to your program so you can actually click the submit button and see a confirmation message.

6. CSS Art Challenge

Project Prompt

Use only HTML and CSS to create a piece of digital art. You'll build shapes, layer them, and style them to form a character, creature, icon, or abstract design.

What You'll Make

A standalone "artboard" webpage featuring a CSS-only illustration made from divs, borders, gradients, and transforms.

Core Skills You'll Practice

- CSS shapes (borders, circles, triangles)
- Positioning (relative, absolute)
- Layering and stacking elements

Creative Twist

Design a game character or creature using nothing but CSS shapes.

Challenge Mode

Animate one part of your artwork using @keyframes (blinking eyes, floating movement, glowing effects).

Additional challenge: Use HTML and CSS to rebuild [this painting](#) by Piet Mondrian.

Legendary ★★ ★ (20 pts)

7. Multi-Page Website

Project Prompt

Build a full website with multiple connected pages. Each page should have its own content but share a consistent layout and theme.

What You'll Make

A 3–5 page website with navigation links, shared styling, and a clear structure across all pages.

Core Skills You'll Practice

- Multi-page navigation
- Shared CSS files
- Page-to-page consistency
- Responsive layout patterns

Creative Twist

Turn the site into a game wiki or lore compendium with pages for characters, locations, items, or story arcs.

Challenge Mode

Use a shared CSS theme and add responsive breakpoints so the entire site adapts smoothly to different screen sizes.

8. Theme Switcher Page

Project Prompt

Create a webpage that lets users switch between different visual themes, like flipping a switch in a settings menu.

What You'll Make

A single page with a toggle button that changes colors, backgrounds, and styles across the entire layout.

Core Skills You'll Practice

- CSS variables
- JavaScript-free theme switching (or minimal JS if you choose)
- Transitions and smooth visual changes

Creative Twist

Add elemental themes such as Fire, Ice, Nature, Shadow, or Tech — each with its own palette and vibe.

Challenge Mode

Use CSS variables + transitions to animate the theme change so it feels polished and intentional.

9. Magazine Layout Clone

Project Prompt

Recreate the layout of a real magazine or blog page. Focus on structure, spacing, and typography to make it feel like a professional publication.

What You'll Make

A visually rich page with columns, images, headings, pull quotes, and article sections.

Core Skills You'll Practice

- CSS Grid for complex layouts
- Typography hierarchy
- Spacing and alignment
- Image placement and cropping

Creative Twist

Turn it into a game magazine or esports site featuring reviews, character spotlights, or tournament coverage.

Challenge Mode

Match the typography and spacing of your reference layout as closely as possible using Grid and careful CSS tuning.

JavaScript

JAVASCRIPT PROJECTS

Starter (15 pts)

10. Digital Fortune Cookie

Project Prompt

Build a digital fortune cookie that reveals a random message every time the user clicks a button. You control the tone — wise, funny, spooky, chaotic, or themed.

What You'll Make

A simple interactive webpage with a button and a message area that updates using JavaScript.

Core Skills You'll Practice

- Arrays
- Random number generation
- DOM manipulation
- Event listeners

Creative Twist

Create themed fortunes: fantasy prophecies, sci-fi transmissions, or comedic “teacher quotes.”

Challenge Mode

Add rarity levels (Common, Rare, Epic) with color-coded styling.

11. Color Changer

Project Prompt

Create a webpage where clicking different buttons changes the background color. Think of it like adjusting the “mood” of the page.

What You'll Make

A color-changing interface with multiple buttons that update the page's style.

Core Skills You'll Practice

- DOM selection
- Event listeners
- Inline styles vs. CSS classes
- Basic UI feedback

Creative Twist

Turn it into a mood selector with labels like Calm, Energetic, Mysterious, or Chaotic.

Challenge Mode

Add a reset button and switch to class toggles for cleaner styling.

12. Simple Calculator

Project Prompt

Build a basic calculator that performs addition, subtraction, multiplication, and division — just like the classic handheld ones.

What You'll Make

A functional calculator with buttons, a display screen, and basic math operations.

Core Skills You'll Practice

- Handling button clicks
- Updating text content
- Writing functions for math operations
- Managing state (current number, operator, result)

Creative Twist

Style it like a retro game UI — pixel fonts, neon borders, or arcade colors.

Challenge Mode

Support keyboard input and decimal numbers.

Power-Up ★★ (20 pts)

13. Quiz App

Project Prompt

Create a quiz that shows questions one at a time, checks answers, and keeps score.

What You'll Make

A multi-question quiz interface with buttons, score tracking, and feedback.

Core Skills You'll Practice

- Arrays of objects
- Conditional logic
- Updating the DOM dynamically
- Tracking score and progress

Creative Twist

Theme the quiz: zombies, space missions, fantasy lore, or anything you love.

Challenge Mode

Add a timer and randomized question order.

14. To-Do List Lite

Project Prompt

Build a simple task manager where users can add and remove items from a list.

What You'll Make

A clean interface with an input box, an "Add" button, and a list of tasks.

Core Skills You'll Practice

- DOM creation and deletion
- Event handling
- Arrays and list management
- Basic UI structure

Creative Twist

Treat tasks as quests — complete them to "level up" your productivity.

Challenge Mode

Save tasks using localStorage so they stay after refresh.

15. Mood Switcher

Project Prompt

Create a page that switches between different themes — like flipping a settings toggle in a game.

What You'll Make

A theme-toggle interface that changes colors, fonts, or backgrounds.

Core Skills You'll Practice

- Class toggling
- DOM updates
- Theme design
- Basic state management

Creative Twist

Add sound effects or small animations when switching themes.

Challenge Mode

Save the selected theme so it loads automatically next time.

Legendary (25 pts)

16. Memory Card Game

Project Prompt

Build a classic memory-matching game where players flip cards to find pairs.

What You'll Make

A grid of cards that flip, match, and reset — all controlled by JavaScript.

Core Skills You'll Practice

- Game logic
- State management
- Timers
- DOM manipulation and animations

Creative Twist

Use your own art, icons, or characters for the cards.

Challenge Mode

Add a timer, scoring system, and difficulty levels.

17. Typing Speed Test

Project Prompt

Create a typing test that measures how fast and accurately a user types a given sentence.

What You'll Make

A typing interface with a prompt, input box, timer, and results screen.

Core Skills You'll Practice

- Timers
- String comparison
- Real-time input tracking
- Calculating words per minute (WPM)

Creative Twist

Use themed sentences: lore entries, character quotes, or sci-fi transmissions.

Challenge Mode

Highlight mistakes in real time as the user types.

18. Drawing App

Project Prompt

Build a simple drawing tool where users can draw on a canvas using their mouse or finger.

What You'll Make

A canvas-based drawing interface with color options and brush controls.

Core Skills You'll Practice

- Canvas API
- Mouse/touch events
- Color and brush size controls
- Clearing and updating the canvas

Creative Twist

Add multiple brush types, color palettes, and an eraser mode.

Challenge Mode

Allow users to save their drawings.

HTML5 Development

HTML5 APP DEVELOPMENT PROJECTS

Starter (20 pts)

19. Form Validator

Project Prompt

Create a form that checks whether user input is valid before submitting. You'll guide users to enter correct information using real-time feedback.

What You'll Make

A character-creation-style form with fields like name, class, stats, or backstory — all validated with JavaScript.

Core Skills You'll Practice

- Form structure (HTML)
- Input validation (JavaScript)
- Showing/hiding error messages
- Styling valid/invalid states with CSS

Creative Twist

Turn the form into a character creation screen for a game.

Challenge Mode

Add custom error messages that match your theme.

20. Interactive Checklist

Project Prompt

Build a checklist where users can add items, check them off, and remove them as needed.

What You'll Make

A dynamic list that updates instantly when users interact with it.

Core Skills You'll Practice

- DOM creation and deletion
- Event listeners and state management
- Basic UI styling

Creative Twist

Transform it into a quest tracker with tasks like "Defeat the Goblin King" or "Collect 10 Herbs."

Challenge Mode

Save progress using localStorage so quests stay completed after refresh.

21. Weather Mood Board

Project Prompt

Create a page where selecting a type of weather displays a matching image, message, or vibe.

What You'll Make

A mood board that updates based on user selection — sunny, rainy, snowy, stormy, etc.

Core Skills You'll Practice

Event handling

Conditional logic

Updating images and text dynamically

CSS styling for themes

Creative Twist

Add weather-based buffs like "+2 Speed in Sunshine" or "+5 Wisdom in Fog."

Challenge Mode

Add animations when the weather changes.

Power-Up ★★ (25 pts)

22. Multi-Step Survey

Project Prompt

Build a form that progresses through multiple steps instead of showing everything at once.

What You'll Make

A multi-screen survey with "Next" and "Back" buttons, plus a final results screen.

Core Skills You'll Practice

- Multi-step UI flow
- Showing/hiding sections
- Form validation
- Progress tracking

Creative Twist

Turn it into a personality quiz with themed results.

Challenge Mode

Add a progress bar and step-by-step validation.

23. Recipe Finder App

Project Prompt

Create an app where users can search for recipes and view matching results.

What You'll Make

A search bar, results list, and recipe cards — all updated dynamically.

Core Skills You'll Practice

- Fetching data
- Rendering lists
- Handling loading states
- Basic API usage

Creative Twist

Make it a potion-crafting app where ingredients generate magical brews.

Challenge Mode

Use a real API and show loading animations or “no results” messages.

24. Habit Tracker

Project Prompt

Build a tool that tracks daily habits and shows progress over time.

What You'll Make

A dashboard with habits, checkboxes, and visual indicators of progress.

Core Skills You'll Practice

- State management
- LocalStorage
- Dynamic UI updates
- Basic data visualization

Creative Twist

Turn habits into XP and levels, like a character progression system.

Challenge Mode

Add streaks and visual indicators for consistency.

Legendary (30 pts)

25. Budget Tracker App

Project Prompt

Create an app where users can add expenses, categorize them, and see totals.

What You'll Make

A dashboard with input fields, a transaction list, and summary totals.

Core Skills You'll Practice

- Data structures (arrays of objects)
- Summing and filtering data
- Dynamic rendering
- LocalStorage

Creative Twist

Style it like an in-game shop ledger tracking gold, items, and purchases.

Challenge Mode

Add charts and categories for deeper insights.

26. Flashcard Study App

Project Prompt

Build an app where users can create flashcards, flip them, and study.

What You'll Make

A card-flipping interface with front/back views and a list of saved cards.

Core Skills You'll Practice

- DOM creation
- CSS transitions
- State management
- LocalStorage

Creative Twist

Add themes like fantasy, sci-fi, or retro arcade.

Challenge Mode

Implement spaced repetition logic to schedule when cards appear.

27. Avatar Builder

Project Prompt

Create a character builder where users customize appearance, gear, or traits.

What You'll Make

A live preview area plus controls for selecting features like hair, colors, outfits, or stats.

Core Skills You'll Practice

- Dynamic styling
- Updating images or SVGs
- State management
- Saving and loading data

Creative Twist

Add stats that change based on user choices (e.g., armor increases Defense).

Challenge Mode

Save the avatar and show a live preview that updates instantly.